

AGENEWTECH

H1641RP

Smart Module Specification_V2.2



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About the Document

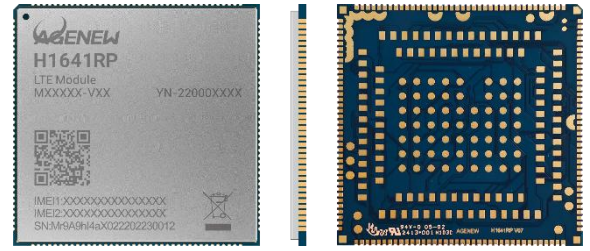
History

Version	Data	Author	Description
-	2022-06-14	Guan Qibin	Initial
1.0	2022-12-04	Guan Qibin	Controlled Copy
1.1	2023-11-30	Peng Cong	Upgrade the software version
2.0	2024-05-05	Wu Danyi	Upgrade to single-sided fabric version
2.1	2025-11-25	Wu Danyi	Add compatible memory information
2.2	2026-03-24	Chen Yun	Change company name

H1641RP

Multi-network Smart 4G module

Supports Wi-Fi & BT



H1641RP is a multi-network Smart LTE Cat 7 module based on MediaTek's Octa-core (ARM Cortex-A75 + ARM Cortex-A55) processor MT8786 launched by Wuxi Sunning Technology Co., Ltd, which is equipped with Android 14 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

H1641RP supports full coverage of various network standards, Wi-Fi 802.11a/b/g/n/ac and BT5.0 LE short-range communication. Not only that, H1641RP also has audio and video I/O interfaces and rich GPIO interfaces, but also supports a variety of audio and video codecs, and the main screen supports up to FHD+ 2520×1080. At the same time, the positioning system of GPS/GLONASS/BDS/Galileo are supported by H1641RP makes its positioning function fast and accurate.

H1641RP is a patch module with 321 pins, including 152 LCC pins and 169 LGA pins. With a size of only 43.0mm×44.0mm×2.6mm, it can be embedded in various M2M products through pads, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle-mounted equipment, high-end information collection equipment, intelligent robots, smart homes, smart hardware, industrial smart handheld equipment, unmanned aerial vehicles, high-end police or law enforcement equipment, smart intercom equipment, smart wearing and vending machines.



Main Advantages

- ✓ Comprehensive coverage of multi-network standard.
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT5.0 short-range wireless communication
- ✓ Support GPS/GLONASS/BDS/Galileo satellite positioning
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to FHD+ 2520×1080
- ✓ Support 1080P@30fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



Dual-core ARM Cortex-A75
+
Six-core ARM Cortex-A55



IEEE 802.11 a/b/g/n/ac



Support Android 14



GPS/GLONASS/BDS/Galileo



Powerful multimedia



BT5.0

LTE Cat7		H1641RP
Region/Operator	China/India/South Korea/Thailand/Japan/Australia/Taiwan/Brazil, Europe, Middle East, Africa, North America. Latin America	
Operating system	Android 14	
Memory	4GB + 64GB / 8GB + 128GB / 8GB + 256GB	
Module size (mm)	43.0× 44.0 ×2.6	
Packaging	LCC + LGA	
Temperature range		
Operation temperature	Operating temperature: -20 °C ~ + 80°C	
	Storage temperature range: -40 °C ~ + 90 °C	
Frequency band information		
LTE-FDD	B1/B3/B5/B7/B8/B20	
LTE-TDD	B34/B38/B39/B40/B41	
WCDMA	B1/B2/B5/ B8	
TD-SCDMA	B34/B39	
GSM/EDGE	B2/B3/B5/ B8	
WLAN	2.4/5GHz, 802.11 a/b/g/n/ac, Support AP and STA mode	
Bluetooth	BT 1.2/2.1+EDR/4.2/5.0	
GNSS	GPS/GLONASS/BDS/Galileo	
Authentication		
Operator Certification	Undetermined	
Compulsory Certification	CE/SRRC*/NAL*/CCC*	
Other Certifications	RoHS/REACH	
Frequency Band Information		
LTE-FDD (Mbps)	300Mbps (DL)/75Mbps (UL)	
LTE-TDD (Mbps)	260Mbps (DL)/90Mbps (UL)	
DC-HSPA+ (Mbps)	42Mbps (DL)/11.2Mbps (UL)	
WCDMA (Kbps)	384Kbps (DL)/384Kbps (UL)	
EDGE (Kbps)	296Kbps (DL)/236.8Kbps (UL)	
GPRS (Kbps)	107Kbps (DL)/85.6Kbps (UL)	
Outstanding characteristics		
Five-mode communication	LTE: Release 12 UMTS: Release 9	
Bandwidth (MHz)	1.4 - 20 MHz Radio frequency bandwidth	
Charging chip	●	
OTG	●	
SIM card detection	●	
Dual sim dual standby	●	
USB Firmware upgrade	●	
OTA Firmware upgrade	●	

Remarks: 1. *: In progress
2. ●: Supports

LTE Cat7 H1641RP	
Electrical specification	
Operating Voltage	3.4~4.4 V, typical value 3.8 V
Consumption (typical)	Undetermined
Interface	
LCD	4-lane MIPI DSI FHD+ 2520×1080
Camera	×3, 4-lane MIPI CSI, up to 25MP
TP	Capacitive touch screen
Audio	3 Analog inputs: 3 groups of analog microphone inputs 3 Analog outputs: speaker, AB-class receiver and AB-class stereo earphone
USB	×1, USB 2.0 (support OTG)
(U)SIM	×2 (1.8V/2.95V)
UART	×2
SD Card	×1
ADC	×1
I2C	×6
SPI	×5
PWRKEY	Internal pull up to VBAT
GPIO	×22
Antenna	Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna